RAG Document Management: Complete Theory & Practice Guide

Mastering the critical gap between demo-ready RAG systems and production-grade document management. A comprehensive framework for engineers building real-world retrieval systems.

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The Demo-to-Production Gap

The Demo Assumption

Classic RAG tutorials show a clean pipeline: chunk documents, embed them, store in vector database, retrieve, and generate. This works beautifully in controlled environments where documents are static, every user sees identical content, and there's one unchanging source of truth.

The Production Reality

The moment you deploy in a real company, chaos emerges. HR updates policies quarterly. Sales uploads 50 pitch deck versions. Users expect private document access. Legal demands audit trails. The CEO questions why outdated strategy docs appear in results.

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Why This Topic Matters

System Reliability

Improper document management causes retrieval failures, inconsistent results, and system crashes when documents update.

Security & Compliance

Multi-user systems require strict data isolation. A single leak violates privacy regulations and destroys user trust.

Operational Costs

Poor duplicate handling and inefficient updates waste compute resources and inflate storage costs exponentially.

User Experience

Stale content, duplicate results, and slow queries drive users away from your RAG system regardless of answer quality.

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PART 1

The Identity Problem

Before managing documents, you must answer a fundamental question: what makes a document a document? Consider this scenario that reveals the complexity beneath the surface.

The Identity Challenge

A user uploads "Budget_2024.xlsx" on Monday. Wednesday brings an updated version with revised numbers, same filename. Friday's "Budget_2024_Final.xlsx" is actually identical to Wednesday's file despite the different name.

01

Are Monday and Wednesday the same document or different?

02

Are Wednesday and Friday the same despite different names?

03

Which version should retrieval return when queried?

04

Should old versions be kept for audit or deleted?

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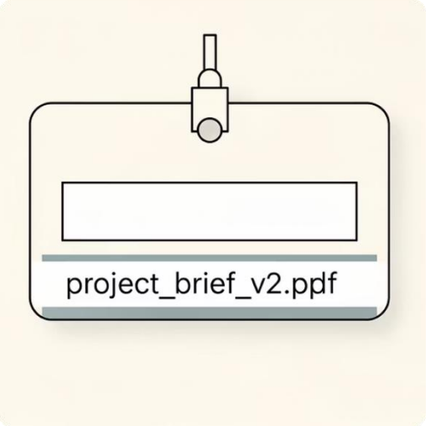
Three Approaches to Document Identity



Name-Based Identity

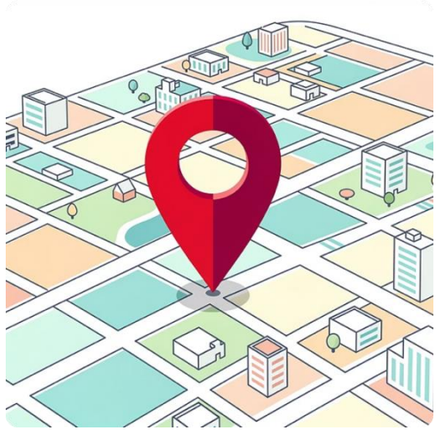
Same filename equals same document. Simple but fragile when users rename files constantly, creating unintended duplicates.

Best for: Strictly controlled naming conventions in internal tools.

Content-Based Identity

SHA-256 hash determines identity. Catches exact duplicates but misses near-duplicates when content changes slightly.

Best for: Detecting exact duplicates and preventing redundant processing.



Source-Based Identity

Identity tied to source location. Stable and robust but doesn't work for user uploads without fixed sources.

Best for: Enterprise deployments with controlled document repositories.

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The Practical Solution: Hybrid Identity

Real production systems don't choose one approach—they combine all three strategically based on document source and use case.

1

Corporate Documents

Identity = source path. Updates detected by content hash. Version history maintained automatically.

2

User Uploads

Identity = user_id + filename + timestamp. Content hash for duplicates. User prompted on near-matches.

3

Hybrid Result

Stability without document multiplication, handling real-world upload messiness gracefully.

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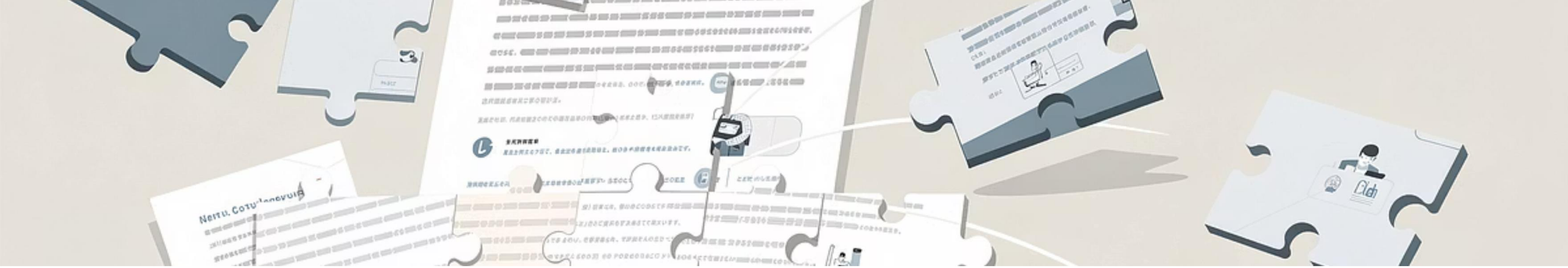

PART 2

The Chunking Versioning Problem

Document identity is just the beginning. When documents update, the real complexity emerges at the chunk level. This problem trips up even experienced engineers.

Why Chunk Identity is Harder

Your Q3 report has 100 chunks stored in the vector database. Finance fixes a single typo on page 3 and re-uploads. Now what happens to those 100 chunks?

Full Re-chunking

Approach: Delete all 100 chunks, create new ones.

Pros: Simple, consistent, no stale data.

Cons: Expensive to re-embed everything when 99 chunks didn't change.

Differential Chunking

Approach: Detect changed sections, only re-chunk those.

Pros: Efficient, faster updates.

Cons: Complex implementation, chunk boundaries may shift creating inconsistencies.

Append-Only Soft Deletes

Approach: Keep old chunks marked "superseded," add new chunks.

Pros: Audit trail, version recovery, no data loss.

Cons: Index bloat, query complexity, storage costs escalate.

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How Companies Actually Handle Updates

1

Startups

Full re-embedding on every update. With corpora under 100K documents, the cost is negligible. Simplicity wins over optimization.

2

Mid-Size Companies

Full re-embedding with queuing. If a document updates 5 times in an hour, only the final version gets processed.

3

Enterprises

Append-only with versioning. Compliance requirements mandate historical version retention. Queries filter to "current version only" by default.

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The Chunk ID Design Decision

Your chunk identifier scheme fundamentally determines how easy updates become. Poor choices create maintenance nightmares.

❌ Sequential IDs

`chunk_001, chunk_002, chunk_003`

Deleting chunk_002 and re-inserting creates gaps. Messy to track and error-prone.

✓ Document-Scoped

`doc123_chunk_001, doc123_chunk_002`

Easier to delete all chunks for a document, but still has gap tracking issues.

✓✓ Version-Aware

`doc123_v1_chunk_001, doc123_v2_chunk_001`

Updates create new version namespace. Old chunks deleted or retained. No conflicts, clean management.

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PART 3

Namespace Theory and Design

Namespaces are the foundation of scalable, secure RAG systems.
Understanding them deeply separates production systems from prototypes.

What is a Namespace, Really?

Think of namespaces as isolated folders inside your vector index. Vectors in different namespaces are completely separated—a query to namespace A will never return results from namespace B, providing true isolation.

Why Not Just Metadata Filtering?

You could store everything in one index and filter by metadata like user_id or department. This works initially but creates problems at scale.

Performance Issues

The database must scan all vectors first, then filter. With millions of vectors, queries become painfully slow.

No True Isolation

A bug in filter logic could leak data across users or tenants. Security depends entirely on application code correctness.

Operational Complexity

Deleting all of Alice's data requires scanning every vector in the database. Namespaces enable atomic operations.

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Namespace Design Patterns

User-Per-Namespace

Each user gets their own namespace: user_alice, user_bob, user_charlie. Perfect for "chat with your documents" apps requiring complete privacy.

Pros: Perfect isolation, easy GDPR deletion, simple mental model.

Cons: Can't search across users, many namespaces to manage.

Tenant-Per-Namespace

Each customer is a tenant: tenant_acme_corp, tenant_globex, tenant_initech. Standard for B2B SaaS applications.

Pros: Strong isolation, per-tenant scaling, compliance-friendly.

Cons: Cross-tenant analytics impossible, namespace management overhead.

Scope-Based Namespaces

Mixed strategy: corporate_hr, corporate_engineering, user_alice, project_moonshot, session_xyz. Flexible hierarchy for complex enterprises.

Pros: Supports multiple access patterns, clear boundaries, maximum flexibility.

Cons: Query routing complexity—which namespaces to search?

The Namespace Routing Problem

Alice is an engineer on Project Moonshot. She asks: "What's the vacation policy?" Which namespaces should your system search?

- **user_alice**

  Her personal uploads—maybe she saved a policy summary.

- **corporate_hr**

  The official policy lives here. Highest relevance.

- **corporate_engineering**

  Her department might have specific rules.

- **project_moonshot**

  Probably not relevant to vacation policy.

- **session_abc**

  Only if she uploaded something this session.

The Solution: Intent-Based Routing

Analyze the query to determine relevant namespaces. "Vacation policy" triggers HR-related routing. "Project timeline" routes to project namespaces. "Document I just uploaded" routes to session only.

Use an LLM classifier or embedding similarity to intelligently route queries to appropriate namespaces.

PART 4

The Multi-Tenancy Challenge

Multi-tenancy means multiple customers sharing infrastructure. In RAG systems, this raises critical questions about isolation, performance, costs, and customization.

Multi-Tenancy Core Questions

Data Isolation

Can Acme Corp's data ever leak to Globex under any circumstances? What guarantees prevent cross-tenant exposure?

Performance Isolation

Does Acme's heavy batch processing slow down Globex's real-time queries? How do you prevent the noisy neighbor problem?

Cost Allocation

How do you fairly bill each tenant when they share infrastructure? Who pays for what?

Customization

Can tenants have different embedding models, chunking strategies, or retention policies while sharing systems?

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Isolation Model 1: Shared Everything

All tenants share one index. Every vector has tenant_id metadata. Queries always include a tenant filter.

The Architecture

Single vector index contains all tenant data mixed together. Application layer filters by tenant_id on every query. Lowest cost, highest risk approach.

Risk: A single bug in filtering logic exposes everyone's data. A query without a filter returns all tenant data.

When to Use

Early-stage startups with non-sensitive data and cost constraints. Acceptable when data leakage wouldn't be catastrophic.

Mitigation: Defense in depth—filter at both application and database layers. Comprehensive audit logs for cross-tenant queries.

Isolation Model 2: Shared Index, Separate Namespaces

One index with each tenant in a dedicated namespace. Queries routed to tenant's namespace. The standard approach for most B2B SaaS applications.

Architecture Benefits

Physical or logical separation within shared infrastructure. Namespace misconfiguration risk exists but is much lower than pure filtering.

Risk Management

Namespace naming convention includes tenant ID. Application layer validates namespace before every query. Resource contention possible between tenants.

Best For

Most B2B SaaS applications. Excellent balance between isolation and cost-efficiency. Industry standard approach.

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Isolation Model 3: Separate Indexes

Each tenant gets their own dedicated index or even database instance. Maximum isolation with maximum operational overhead.

1

True Isolation

Each tenant has completely separate infrastructure. No shared resources, no cross-tenant risk whatsoever.

2

The Cost

Operational overhead multiplies with tenant count. Costs scale linearly. Requires extensive automation for index provisioning.

3

When Required

Enterprise customers with compliance requirements, high-value tenants who pay for isolation, regulated industries like healthcare and finance.

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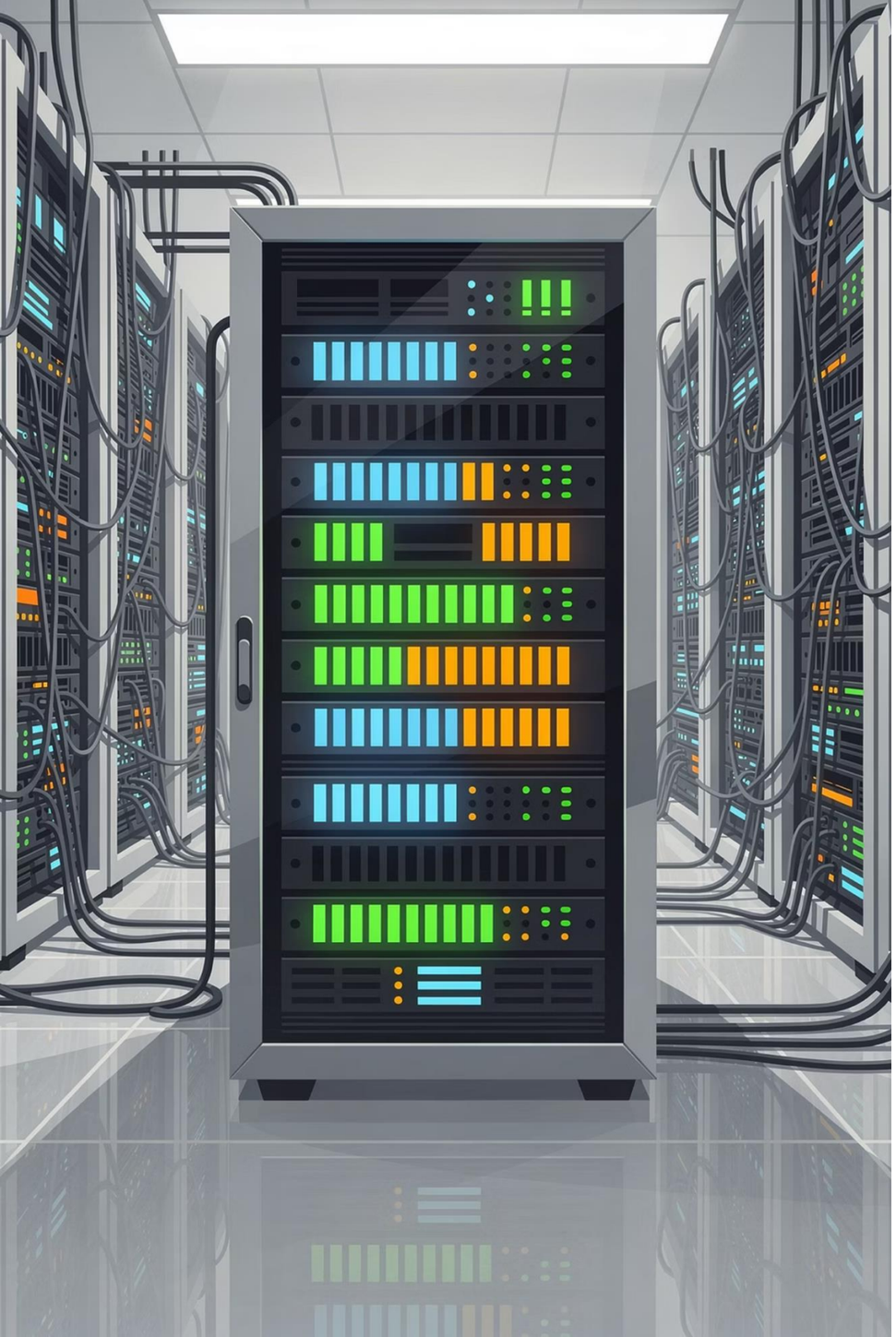

The Noisy Neighbor Problem

Even with perfect data isolation, tenants share compute resources. What happens when Acme runs a massive batch job while Globex's CEO demos the product to their board?

Solving the Noisy Neighbor Problem

Rate Limiting Per Tenant

Each tenant gets maximum requests per minute. Acme's batch job gets throttled, protecting other tenants.

Priority Queues

Real-time queries go to high-priority queue. Batch jobs process in background queue with lower priority.

Tenant Tiers

Enterprise tenants get dedicated capacity. Smaller tenants share pooled resources. Fair allocation based on payment.

Separate Infrastructure

Biggest customers get their own dedicated clusters, completely isolated from other tenants.

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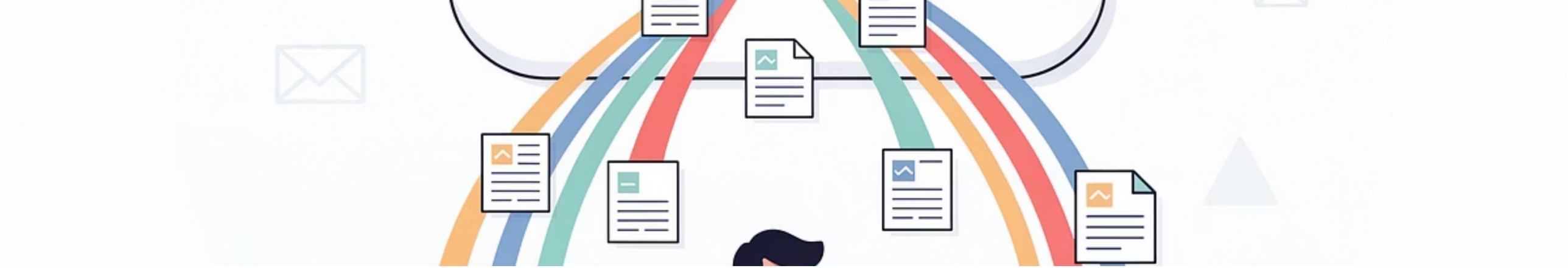

PART 5

User Upload Scenarios Deep Dive

Different use cases require radically different architectures. Understanding these scenarios helps you design systems that match user expectations.

Scenario 1: Chat with PDF - Single Session

The simplest case: user uploads a PDF, asks questions, then leaves. They don't expect to see this PDF tomorrow.

User Mental Model

"I'm uploading this for this conversation only. It's like showing a document to a friend—they don't keep a copy."

System Design

Create session-scoped namespace (session_xyz123). Embed PDF chunks into this namespace. On query, search only this namespace. On session end, delete namespace entirely—all vectors gone.

TTL Strategy: 24 hours of inactivity triggers auto-delete. Explicit session end causes immediate deletion. Storage quota reached deletes oldest sessions.

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Why Namespaces for Sessions?

You could use metadata filtering with session_id, but namespaces provide critical advantages that protect your system and users.

Guaranteed Isolation

No filter bugs can leak one session's documents into another. Physical separation ensures privacy.

Atomic Cleanup

Delete namespace vs. delete-by-filter query. Much faster and more reliable.

Clear Boundaries

Easy to count vectors per session, enforce limits, monitor resource usage.

Scenario 2: My Document Library - Persistent Storage

Users upload documents to their personal library. They expect access across sessions, indefinitely. This requires different architectural decisions.

User Mental Model

"This is my folder in the cloud. I can upload documents, organize them, and ask questions across all my documents anytime."

Namespace Strategy

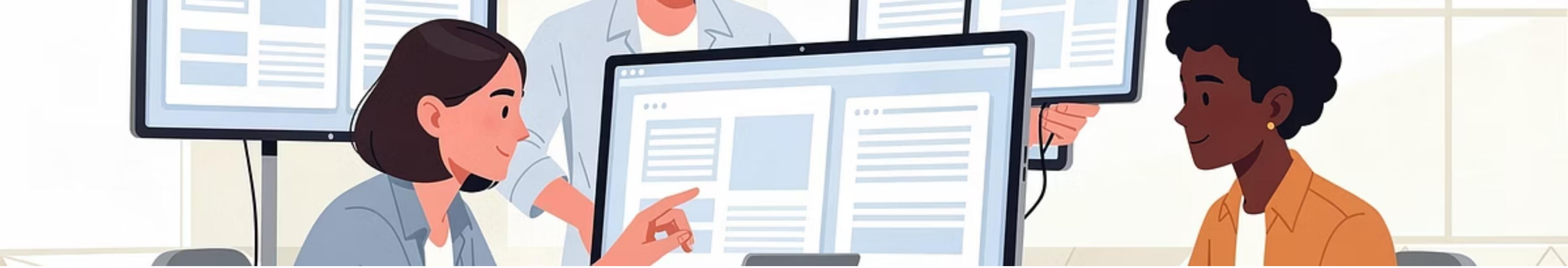
user_{user_id} namespace persists indefinitely. On upload, check for duplicates by content hash. If new, embed and store. If duplicate, skip or ask user.

Storage Management

Enforce quotas: 100 documents or 500MB limits. Provide document management UI for users to see library contents, delete old docs, organize folders.

Search Scope

Support "search my documents" versus "search everything I have access to" including shared corporate docs.



Scenario 3: Team/Project Shared Documents

A group collaborates on a project. They share documents that all team members can query, adding complexity around access control and ownership.

Shared Document Complications

The Challenge

Alice uploads "roadmap.pdf" on Monday. Bob uploads "roadmap.pdf" with different content on Tuesday. Which one wins?

Options include last-write-wins (Bob's version), version both as roadmap_v1 and roadmap_v2, or prompt uploader about the conflict.

Architecture Decisions

Namespace: project_{project_id} with access control by membership.

Uploads: Any member can add, but track who uploaded what for audit purposes.

Member Removal: They lose access but their uploads remain, owned by project not person.

Scenario 4: Corporate Knowledge Base + User Uploads

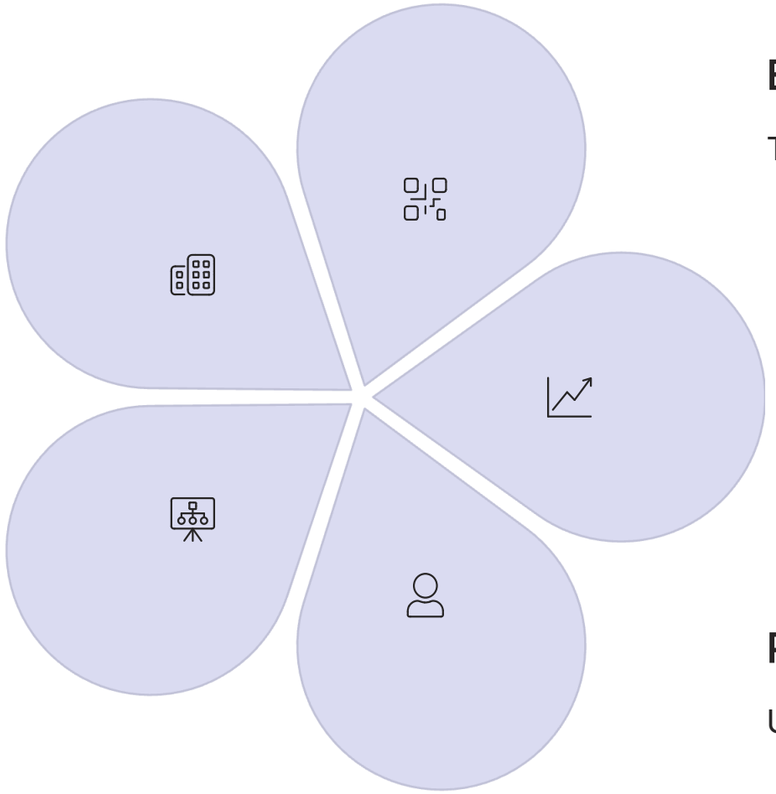
The most complex scenario: official company documents plus personal user uploads. Queries should intelligently search across both while maintaining security boundaries.

Corporate HR

Official policies and benefits

Engineering Docs

Technical documentation



Sales Materials

Pitch decks and case studies

Project Spaces

Cross-functional collaboration

Personal Uploads

User-specific documents

Queries must route to relevant namespaces, merge results, deduplicate, and rerank by relevance while respecting access permissions.

Query Routing Architecture

For complex environments like Scenario 4, where corporate knowledge coexists with user-specific uploads, an intelligent query routing architecture is essential. This system dynamically processes each query to ensure security, relevance, and a seamless user experience.

User Query & Context

Alice (Engineer on Project Apollo) submits a search query. The system captures her user ID and contextual information.

Access Control & Permissions

The system instantly determines Alice's access permissions across various namespaces: Corporate HR, Corporate Engineering, User Alice's personal files, and Project Apollo's shared documents.

Parallel Search

Simultaneous, secure searches are initiated across all accessible namespaces. This ensures comprehensive coverage while respecting data boundaries.

Result Aggregation

Results from all parallel searches are merged, deduplicated to remove redundant entries, and then intelligently reranked based on relevance to the query and Alice's context.

Unified Results

Alice receives a single, coherent set of top results, drawn from all authorized sources in real-time, providing a complete picture without exposing sensitive data.

This sophisticated, real-time routing ensures that every query delivers accurate, relevant, and securely filtered information, critical for complex enterprise knowledge bases.

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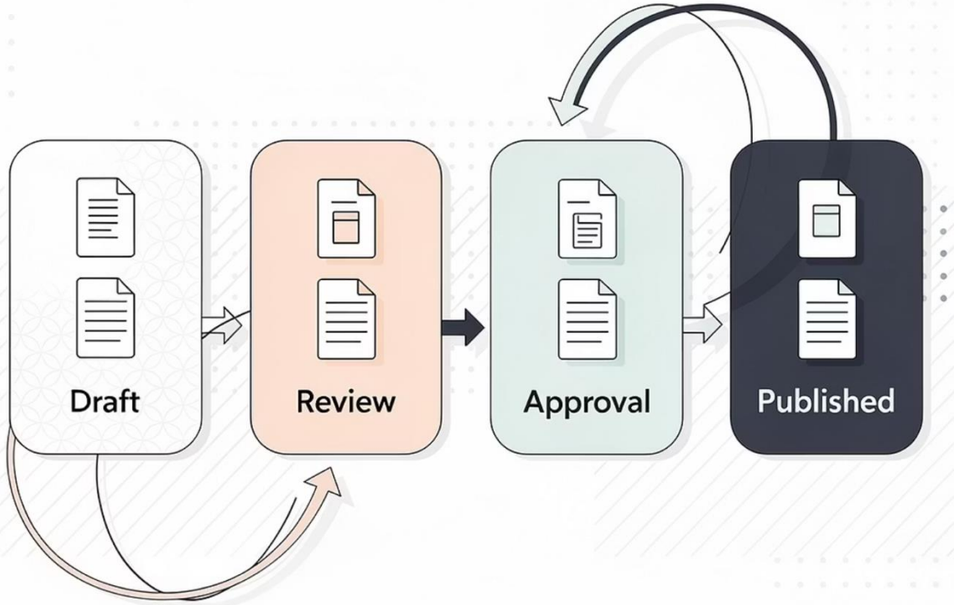
Critical Safeguard: Namespace Isolation

If Alice uploads a document called "Leave Policy.pdf" with her personal notes, it must go to user_alice namespace. It should never appear when Bob asks about leave policy—that should come from corporate_hr.

Why namespace separation is critical: Metadata filtering in a shared index risks pollution. A user's personal document could accidentally influence corporate search results. Namespaces provide hard boundaries that prevent this contamination entirely.

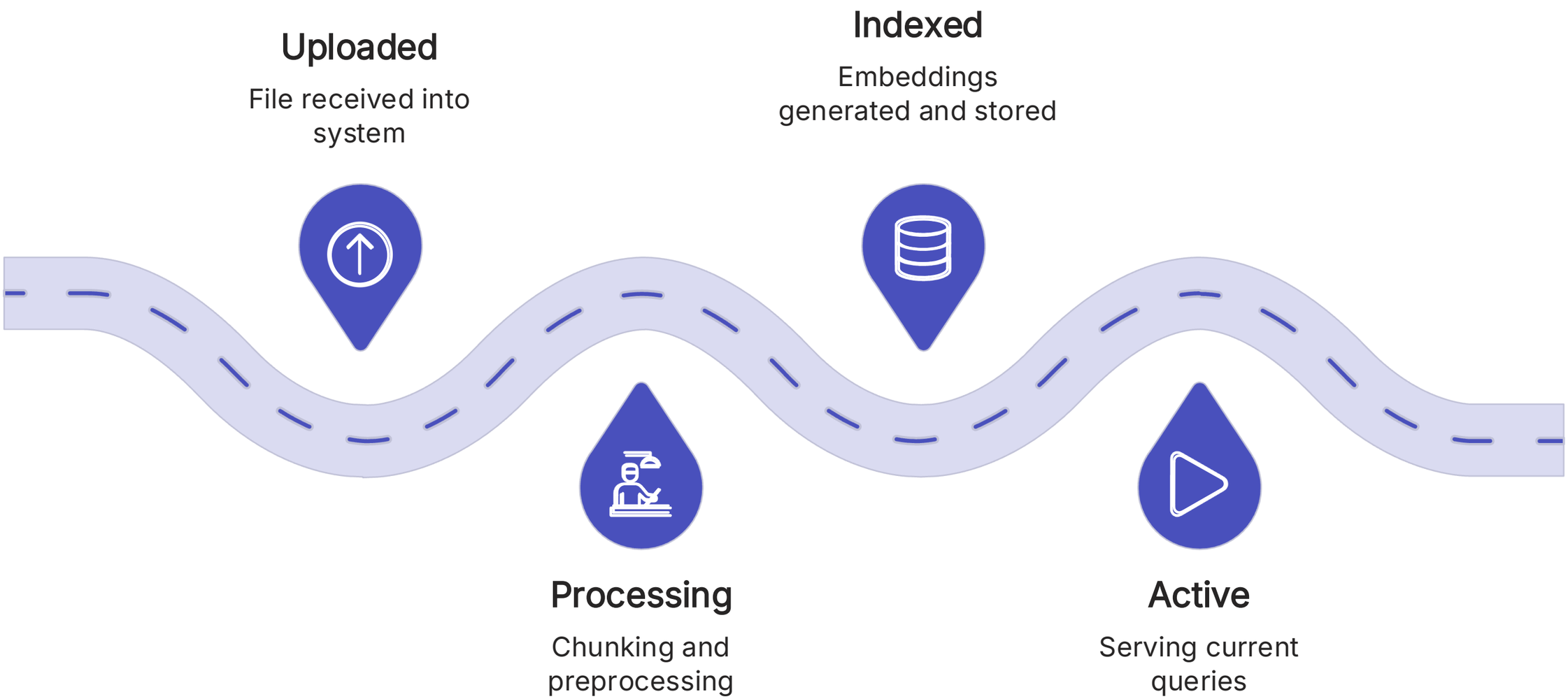


PART 6

Document Lifecycle States

Documents aren't simply "present" or "absent." Production systems require sophisticated state management to handle updates, compliance, and user expectations.

Complete Document Lifecycle



State transitions must be atomic and consistent between document store and vector store to prevent inconsistencies.

State Definitions

State	Meaning	Searchable?	Stored?
UPLOADED	File received, not yet processed	No	File only
PROCESSING	Being chunked/embedded	No	File + partial vectors
INDEXED	Embedding complete	Yes	File + vectors
ACTIVE	Current version, serving queries	Yes	File + vectors
SUPERSEDED	Newer version exists	Configurable	File + vectors (maybe)
ARCHIVED	Removed from search, kept for compliance	No	File + vectors
DELETED	Marked for deletion	No	Maybe file, no vectors
PURGED	Completely removed	No	Nothing

Why State Matters: Compliance Scenario

Compliance team needs to find all documents mentioning "Project X" from 2023, including deleted ones.

With True Deletion

If you PURGED documents, this request is impossible. Data is gone forever. Compliance violations, audit failures, and legal exposure result.

With Soft Deletion

If you ARCHIVED documents, you can search the archive. Compliance requirements met, audit trail preserved, legal discovery supported.

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State Transition Best Practices

The key insight: state transitions must be atomic and consistent between your document store (Postgres) and vector store (Pinecone).

❌ Bad Approach

1. Delete old vectors from Pinecone
2. Upload new vectors to Pinecone
3. Update status in Postgres

If step 2 fails, you have no vectors but Postgres thinks you do. System inconsistent.

✓ Better Approach

1. Upload new vectors with status="pending"
2. Update Postgres: old → SUPERSEDED, new → ACTIVE
3. Delete old vectors
4. Update new vectors: status="active"

If any step fails, you can recover gracefully. No user-visible inconsistency.

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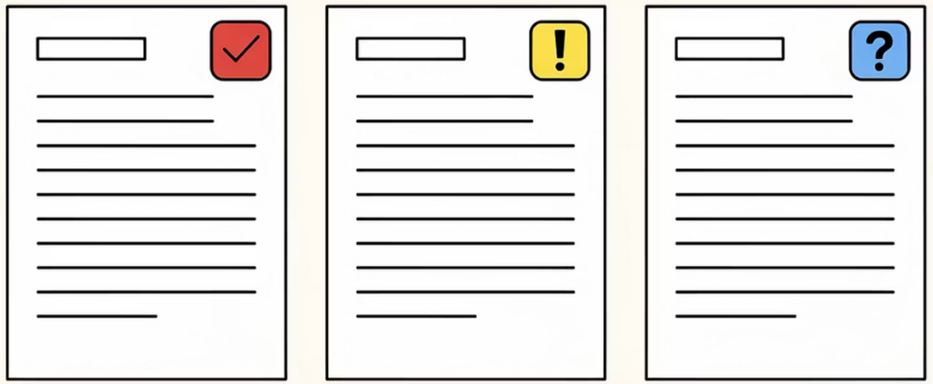
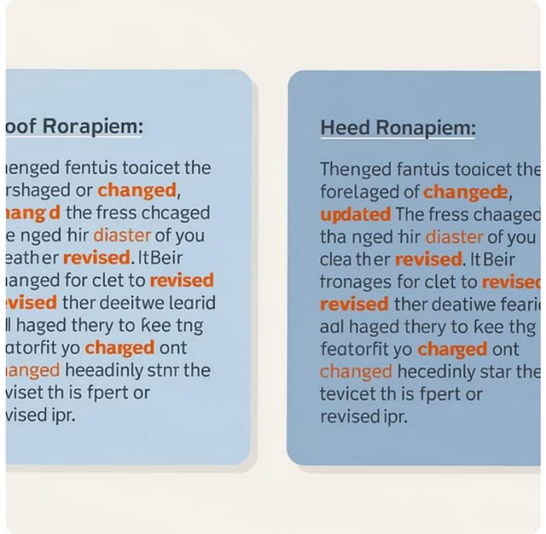


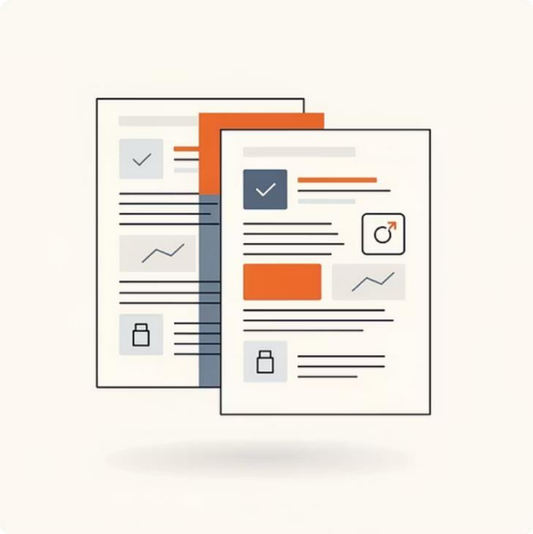
PART 7

Duplicate Detection Strategies

Detecting duplicates saves compute costs, storage space, and prevents user confusion. But there are three distinct types requiring different detection methods.

The Three Types of Duplicates





Exact Duplicates

Byte-for-byte identical files. Same content, possibly different filenames.

Detection: Content hash (SHA-256). If hashes match, files are identical.

Action: Skip processing, create reference to existing document.

Near Duplicates

Same content with minor variations—different formatting, extra whitespace, OCR errors.

Detection: Similarity of document embeddings. Embed full document, compare to existing.

Action: Alert user, ask whether to replace or keep both.

Partial Duplicates

Documents sharing sections. Slide decks reusing slides, reports quoting other reports.

Detection: Chunk-level similarity. Find chunks similar to existing chunks.

Action: Context-dependent—acceptable for citations, problematic for plagiarism.

Duplicate Detection at Scale

Checking a new document against millions of existing documents is expensive. You need efficient strategies that balance accuracy with performance.

Hash-Based Exact Match

O(1) lookup time. Store content hashes in database. Check new document's hash instantly. Document arrives → Calculate SHA-256 → Check hash table → Match? Skip : Proceed.

LSH for Near Duplicates

Locality Sensitive Hashing buckets similar documents. Instead of O(n) comparisons, check only documents in same bucket. Trade-off: some false negatives but much faster.

First-Chunk Sampling

Don't embed whole document for duplicate detection. Embed just first chunk or few sampled chunks. Compare against first chunks of existing documents.

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Handling Detected Duplicates

What do you do when you find a duplicate? The answer depends on the duplicate type and your use case.

For Exact Duplicates

Don't re-embed—waste of compute. Create reference to existing document. Track that this user also "has" this document for their searches.

For Near Duplicates

Alert user: "This document is very similar to [existing]. Did you mean to update it?" Offer options: Replace existing, Keep both, or Cancel upload.

For Partial Duplicates

Usually index normally. Maybe flag for review if overlap exceeds 80%. Let retrieval and reranker handle redundancy naturally.

PART 8

Real-World Case Studies

Theory meets practice. These case studies show how different companies solve document management problems based on their unique constraints and requirements.

Case Study 1: ChatPDF-Style Application

The Product

Users upload PDFs, ask questions, get answers. Simple consumer application.

Scale: 100K users, average 5 PDFs each, 3 queries per session.

Architecture Choices

Namespace: Session-based (session_{id}). PDFs deleted after 24 hours.

Duplicate handling: None. Each upload is independent, even if 1000 users upload same PDF.

Rationale: Simplicity over optimization. Cross-user deduplication adds complexity with privacy implications. Cost of redundant embedding is acceptable.

Case Study 2: Enterprise Knowledge Management

Company-wide search across all internal documents from multiple sources. Confluence, Google Drive, SharePoint, and Notion all indexed together.

50K

Employees

Active users across organization

10M

Documents

Total indexed content

100K

Queries

Daily search volume

Enterprise Knowledge Management: Architecture

Namespace Strategy

Source-based: confluence, gdrive, sharepoint, notion with metadata for department/team. Each source has different update patterns, permissions, and connectors.

Duplicate Handling

Aggressive deduplication. Same document exists in multiple systems. Use content hash to dedupe. Store once, reference from multiple sources.

Update Strategy

Webhook/polling from each source. When document changes: calculate new hash, if changed re-chunk and re-embed, queue for off-peak processing.

Access Control

Post-retrieval filtering. Retrieve top 50 results, filter to what user can access, return top 10. Source systems have complex permissions impossible to pre-filter.

Case Study 3: Multi-Tenant SaaS for Legal

Law firms upload case documents, contracts, and legal research. Each firm is a tenant with highly sensitive data requiring maximum isolation.

Legal SaaS: Security-First Architecture

Scale & Requirements

Tenants: 500 law firms

Documents: Average 100K per firm

Sensitivity: Highly confidential—any cross-tenant leak is catastrophic

Critical Decisions

Namespace: Tenant-per-namespace (tenant_{firm_id}) with strict isolation. Top 10 largest firms get dedicated indexes.

Duplicates: Within-tenant only. No cross-tenant deduplication despite storage costs.

Updates: Versioned. Legal documents have versions that matter for court proceedings.

Retention: Never delete, only archive. Courts can request documents from years ago.

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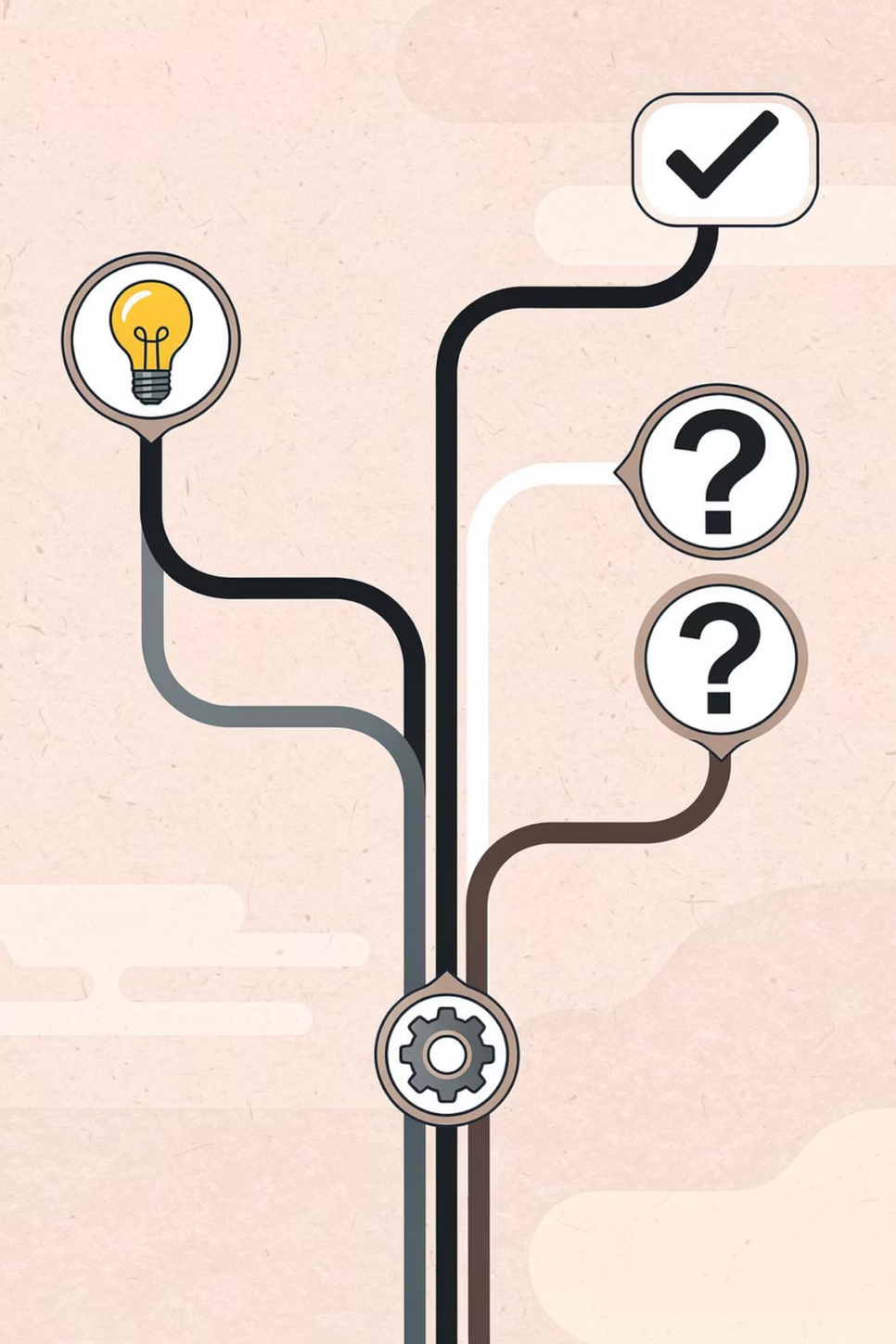
PART 9

Decision Framework for Your Implementation

Choosing the right architecture requires answering specific questions about your use case. Use this framework to guide your decisions.



Choosing a Namespace Strategy

Who owns the data?

Individual users → user namespaces. Teams → project namespaces. Company → corporate namespaces. External customers → tenant namespaces.

What's the isolation requirement?

"Nice to have" → metadata filtering OK. "Must have" → namespaces. "Legally required" → separate indexes.

How is data shared?

Never shared → simple namespace per owner. Shared within groups → namespace per group with query routing. Complex sharing → post-retrieval permission filtering.

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Choosing an Update Strategy

1

Update Frequency

Rarely → re-embed on every change (simple). Frequently → queue updates, batch process. Constantly → consider not using RAG for this data.

2

Version History

Not needed → replace old chunks with new. Required → version-aware chunk IDs, keep old versions for audit and compliance.

3

Staleness Tolerance

Real-time required → synchronous embedding, complex. Minutes OK → async queue, much simpler. Hours OK → batch processing, cheapest.

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Choosing a Duplicate Strategy

Cost of Duplicates

- Just storage/compute → maybe accept some duplication
- User confusion → need deduplication
- Legal/compliance → mandatory deduplication with audit trail

Cost of False Positives

- Accidentally merging different docs → very bad, be conservative with thresholds
- Missing a duplicate → annoying but recoverable

Scope Decision

Within-user only → user-scoped hash tables. Cross-user → global deduplication with privacy implications.

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KEY TAKEAWAYS

Summary: Production RAG Essentials

The Demo-to-Production Gap is Real

Classic RAG assumes static documents. Production has constant updates, duplicates, and multi-tenancy. Understanding document lifecycle separates POCs from production systems.

Document Identity is Foundational

Decide early: name-based, content-based, or source-based identity. Each has trade-offs. Most production systems use hybrid approaches combining all three strategically.

Namespaces Provide Isolation

Use namespaces for security boundaries, not just organization. Metadata filtering is not a substitute for true isolation in multi-user or multi-tenant systems.

State Management Prevents Inconsistency

Documents have lifecycles. Atomic state transitions keep your index consistent between vector store and document store. Design for failure recovery.

Duplicate Detection Saves Money

Use content hashing for exact matches, embedding similarity for near-duplicates. Balance detection cost against storage and compute savings.

Access Control is Post-Retrieval

Pre-filtering is too complex when source systems have intricate permissions. Retrieve broadly, then filter results based on user permissions.

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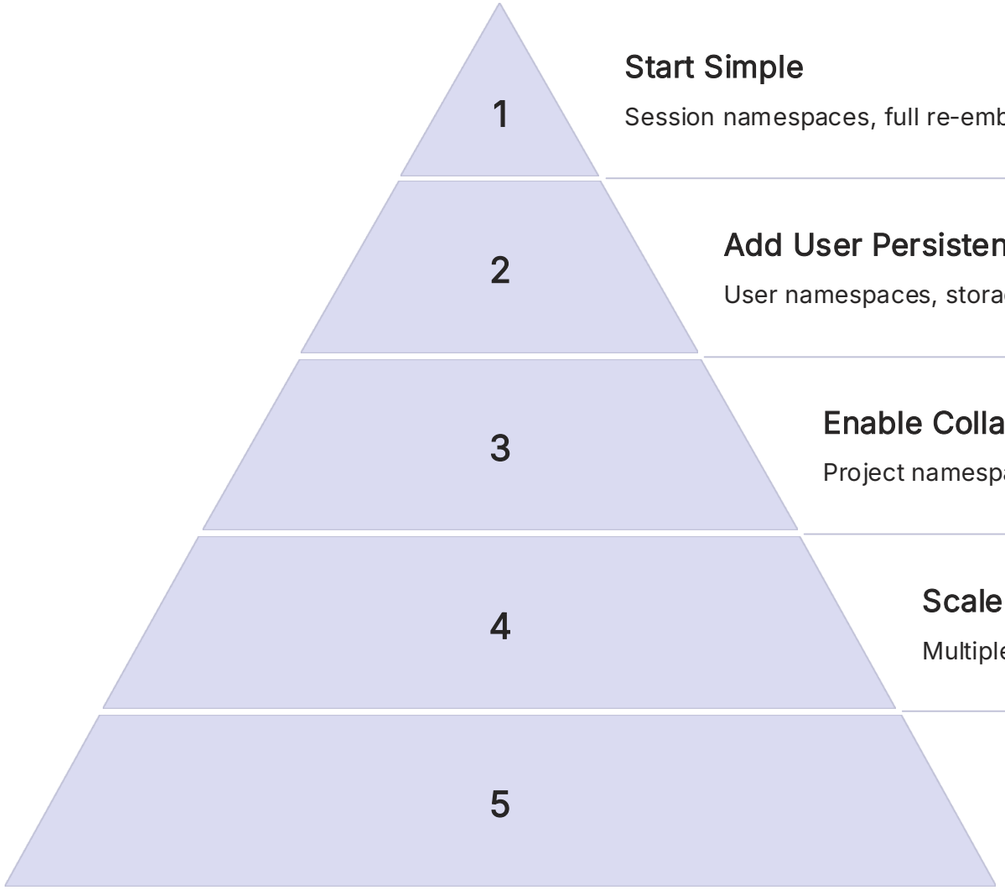
Design for Your Scale

The architecture that works for ChatPDF won't work for enterprise knowledge management. Match complexity to requirements, and don't over-engineer for demos.



Start Simple

Session namespaces, full re-embedding, basic deduplication

Add User Persistence

User namespaces, storage quotas, document management UI

Enable Collaboration

Project namespaces, shared documents, access control

Scale to Enterprise

Multiple sources, complex permissions, versioning, compliance

Multi-Tenant SaaS

Tenant isolation, noisy neighbor prevention, separate indexes for top customers, comprehensive audit trails

Build what you need today, architect for what you'll need tomorrow. Understanding these principles enables you to make informed trade-offs as your system evolves from prototype to production.

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